

I've Hung You From the Moon — 3D Game Production Brief

A complete build specification for a playable 3D browser game, written to be handed to another Claude model.

0. Your Assignment

You are building “**I've Hung You From the Moon,**” a playable 3D narrative-adventure game in **one self-contained HTML file**. It adapts a surreal, rhyming, darkly comic short story about Christopher Carrington, a Wisconsin father who inherits a mansion in San Antonio from his uncle, the Gronfiser, and slowly unravels. Priorities, in order:

1. **Fidelity to the story.** Every named character, object, number, room and line of dialogue listed here is canon. Do not rename, drop or “fix” them (see Section 10).
2. **Visual richness and detail.** Each chapter must look like its own hand-built diorama with distinct lighting and palette.
3. **Playability.** Every chapter has a real mechanic, not just a cutscene. The game should take 25–40 minutes.
4. **Creative embellishment that serves the tone.** Add props, jokes, sound and mini-mechanics, but never contradict the story.

Deliver one `.html` file. If you run short on output, finish the chapters in order and make each one complete rather than leaving all of them thin. A working vertical slice (Chapters 1–3) is better than a broken full game. Include a **chapter-select debug menu** (press the backtick key or use a title-screen link) so every chapter can be tested individually.

1. Vision and Tone

Logline: A man who thinks he is the emperor of his household drives his family south to claim a fortune, and the house, the moon and his own memories keep trying to teach him what he refuses to see.

Tone: Think a Wes Anderson dollhouse crossed with a fever dream and a Dr. Seuss limerick. The comedy is broad and sincere (a man destroys a TV with a broom handle during *The Sound of Music*). Underneath it sits real sadness: a dead father who never spoke, a mother who fled, a wife who is tired, a daughter turning thirteen whose birthday

he ignores. **Play the laughs straight and let the grief sneak up.** The emotional arc is Christopher moving from bluster to the edge of understanding, then sliding back into obsession (the melting spree) at the end.

Central motifs. Recur them constantly:

- **The Moon.** Milky, enormous, and eventually *with a green-and-blue eye*. The title means the family hangs from it. It watches every chapter.
- **Fleas of the mind.** Christopher's intrusive impulses, which hijack his "thought train." Rendered as tiny glowing sparks that crawl across the screen.
- **Rhyme.** Narration, signs and many lines rhyme. The more manic Christopher gets, the more everything rhymes.
- **Gold, birds, racoons, kites and fish.** Each owns a chapter.
- **Chandeliers.** They are everywhere, and their color tells you which reality you are in.

2. Technical Foundation

- **Engine:** Three.js, loaded from cdnjs with a pinned version:
<https://cdnjs.cloudflare.com/ajax/libs/three.js/r128/three.min.js>. No other external scripts, textures, models, fonts or audio files. **Everything is procedural:** geometry from primitives, textures drawn on `<canvas>`, sound from the Web Audio API. Use system font stacks (Georgia and Palatino for narration; monospace for the HUD).
- **Do not rely on `three/examples` (`EffectComposer`, `OrbitControls`, `GLTFLoader`).** They are not in the core build. For post-processing, write your own: render to a `WebGLRenderTarget`, then draw a full-screen quad with a custom fragment shader (vignette, film grain, chromatic aberration, soft bloom approximation, color grade). Provide a quality toggle (High / Low) that disables post-processing and shadows.
- **Architecture:** a `Game` object with a **SceneManager** (each chapter is a class with `build()`, `enter()`, `update(dt)`, `exit()`, `dispose()`), plus `Input` (keyboard, pointer-lock mouse, touch fallback), `Audio`, `UI` (DOM overlay), `Dialogue` and `State` modules. Fixed 60 Hz logic with variable render. Dispose geometries and materials on scene exit.
- **Performance:** target 60 fps on a mid laptop. Use instancing for repeated props (armor parts, candles, pillows, birds, trees, fish), cap shadow-casting lights at 2 per scene, and use a shadow map of 1024–2048.

- **Cameras:** first person by default (visible hands and forearms). Cutscenes and a few set pieces (the road, the lake) use scripted or fixed third-person cameras. Cinematic letterbox bars slide in for cutscenes.
- **Controls:** WASD move, mouse look, **E** interact, **Space** jump or context action, **Shift** sprint, **Q** open the Treasure Notepad, **Esc** pause. Show on-screen prompts. Provide touch controls (virtual stick plus tap-to-interact) as a lower-priority fallback.
- **Save:** hold progress in memory only (do not use localStorage). The chapter-select is the “save” system.

3. Core Systems

Narration voice. A bottom-center caption box in a storybook font. The narrator is Christopher in first person, speaking in rhyming couplets whenever the Flea Meter is above 30%. Dialogue uses a typewriter effect with click-to-skip. Where the story provides lines, **use them verbatim or lightly trimmed.**

Meter 1: The Flea Meter (0–100). It rises when Christopher indulges impulses (smashing, throwing, drinking, melting, ignoring family) and drops when he does tender things (listening, hugging, fishing patiently, singing with the family). Effects scale with the meter: glowing flea sparks crawl along the screen edges; the vignette pulses; a buzzing whisper track grows; text rhymes more; at 70%+ gold dollar-sign motes swirl around the camera and the FOV breathes. It is displayed as a small glass jar of fleas on the HUD.

Meter 2: The Household Heart (4 hearts). Elsa Ray, Mary Sue, Frank and Christopher’s own. Choices fill or drain them. This changes dialogue flavor and epilogue lines, **but never the main plot beats.** The story’s canonical ending is fixed.

The Treasure Notepad. In Chapter 10, Christopher tallies his assets (his compulsion from Wisconsin). Make this a persistent notepad UI, handwritten-looking, listing every item he “owns” as he finds it. It is also the main collectible system and carries into the finale.

Interaction. Every interactable has a highlight outline (a back-face scaled shell shader), a prompt, and a small response. **Add humor to many interactions** (examining a gold bird: “Seventeen thousand dollars, give or take a feather.”).

Choices. Most choice moments are binary or ternary, and *the in-story option is always available and is the one that advances canon.* Alternative choices produce comedy or consequences (extra Flea Meter) and then funnel back to canon.

4. Visual Direction

Build a consistent look: **handcrafted miniature, painterly, slightly warped.**

- **Materials:** MeshStandardMaterial with canvas-generated textures for wood grain, brick, damask wallpaper, marble, red carpet, tile, fabric weave and gold. Add subtle noise to everything so no surface is flat. Gold is `metalness 1, roughness 0.25` with a generated environment cubemap (render a simple gradient-and-light-blobs scene through a `CubeCamera`) so the metal actually reflects something.
- **Lighting:** one hero light per scene plus warm practicals. Use flickering `PointLight` s for candles and fire; `SpotLight` for moon beams and TV glow; a hemisphere light for ambience. Fog in every scene, color-matched to the palette. Light shafts are additive translucent cones with a gradient texture.
- **Post:** vignette, film grain, mild chromatic aberration at the screen edges that increases with the Flea Meter, gentle bloom on emissives, and per-chapter color grading.
- **Particles:** dust motes in sunbeams, fireflies, embers, feathers, snow, gold coins, fleas. Use `Points` with a soft round sprite drawn on canvas.
- **Moon:** a large sphere with a procedural cratered texture (layered noise) and a faint Fresnel glow. Its eye is a plane with a canvas-drawn iris (green and blue radial gradient, a slit pupil, a wet specular highlight) that blinks, tracks the player, and dilates when Christopher lies. It must look unsettlingly alive.
- **Palette per chapter:** Wisconsin living room (cold blue night, warm lamp amber); the Road (inky black, white headlight cones, raccoon-eye reflections); San Antonio dawn (peach, rose, dusty gold); Dome (silver-milk, ink blue); Bird Lantern Hall (green-gold, dirt-brown, twelve warm nests); Maroon Hallway (ruby-maroon); Kitchen (nostalgic 1990s yellow-beige with tacky marble); Lake (violet dusk, glassy teal); Morning Dining Room (honey sunlight, cream); Gallery Rooms (gold, black, red); Smelting Fire (orange, whiskey amber, black trees).

5. Audio (all procedural via Web Audio)

- **Music:** a warm, wobbling, slightly out-of-tune music-box and pump-organ theme in a minor-key waltz. Each chapter has a variation (road: driving pulse with chugging bass; lake: sparse, near-silent chords; melting: frantic circus rhythm). The *Sound of Music* TV scene uses an invented cheery hills-are-alive-style jingle. **Do not reproduce any real song.**

- **SFX:** boot-kick thump, broom handle through glass (CRT pop and fizz), splash, engine, tire hiss, bird chirps (swept sine blips), wind gust (filtered noise), vase shatter, chandelier chimes, axe chops, fire crackle, glass clink, breathy flute notes with vibrato, bobber plops, fish thrash, grackle call, cake-knife scrape.
- **Voices:** no recorded speech. Use per-character “blip” voices (unique pitch and timbre) synced to the typewriter text.
- **Spatial audio:** use `PannerNode` for birds, fire and moon hums. The flea voice is a quiet bandpassed noise buzz that gets louder with the Flea Meter. Provide a mute and a volume slider.

6. Characters (low-poly, stylized, expressive, built from primitives)

- **Christopher Carrington:** first-person protagonist. In cutscenes: gaunt, wild-haired, barrel-chested, a wrinkled shirt that grows sweatier, sootier and more disheveled across the game.
- **Elsa Ray:** wife. Blue eyes. Warm early on, then thinner and tired, hair darker and duller in the kitchen transformation. Her glare is a signature animation: a half-second flash only the player catches.
- **Mary Sue:** daughter, turning **thirteen**. Gold-blond hair. Wears a gold-speckled cone party hat in the kitchen. Briefly *aged* to look like her mother in the Birthday Kitchen.
- **Frank:** young son. Stern, still, a slyness in his smile (Clement’s face). Mature beyond his years in the dream chapters.
- **The Sweaty Man:** barefoot, long scraggly beard, bathrobe, soaked in sweat, a very long finger. A shimmer of steam rises from him.
- **The Errante (the moon):** speaks only as a glowing moon with a green-and-blue eye. Voice blips: low, warm, resonant.
- **Clement Carrington:** Christopher’s dead father. Sits motionless in a canoe. Face half-lit, gaze on the moon.
- **Edith:** Christopher’s mother, seen only in a memory flash (silhouette, sobbing, driving a truck).
- **The Birds:** twelve nest-dwelling birds plus the kite flock. Distinct species by color. They speak in unison in italic rhyme.
- **Gold statues:** Joshua Primo (8 ft, elephant staff), Crumpis (8 ft, baboon), Crumpalump (8 ft, tiger), Curt (3 ft, loon).
- **Racoon:** the night-road racoon, recurring, with reflective eyes.

7. Chapters (in order)

Chapter 1 — “The Fourteenth of September” (Wisconsin living room)

Setup: a snowy Wisconsin night. A cluttered, cozy 1990s living room: floral couch, wood-paneled walls, family photos, a **27-inch RCA CRT television** playing the annual rerun of *The Sound of Music* (a pastiche only: silhouettes spinning on a green hill, no real likenesses). Elsa Ray, Mary Sue and Frank sit hypnotized on the couch. Mary Sue's fish tank sits on a side table. Snow outside, a sleeping highway nearby. **Beats:**

1. The player has free-look and can examine things (a tutorial) for about 30 seconds.
2. The front door bursts open: **the Sweaty Man**, barefoot, in a bathrobe, dripping. Steam and snow melt around him. Christopher's auto-line: he is *about to say* “*listen here pipsqueak!*” A finger presses to the player's lips. “**Relax, I come bearing gifts.**”
3. He produces a **slightly damp letter** from his sleeve, “baking like the sun.” The player rips it open (mouse-drag tear animation).
4. **The Letter.** Display it full-screen on scrolling parchment, with the rhyme *verbatim* from the story (“How do you do? You are my nephew...” through “...a good health”). It names Joshua Primo, the Gronfiser, and the thirteen years of unanswered letters.
5. **Mechanic:** a timed boot-kick swing at the man's backside (“shoo the old hoot with a kick of my boot”). He runs off into the night in his “smelly bath suit.”
6. **The Broom.** Christopher fetches the **yam** and the **grime fighter solution**, sets them on the kitchen floor, and grabs a **broom** from the closet by its handle. Back in the front room, the player **thrusts the handle into the TV** twice. The first stops the song flat and goes black; the second topples it: “*the Carrington family 27” RCA CRT was no more.*” The family's hypnotized stares break.
7. The line: “**And that was the end of my life as I knew it.**” Flea Meter +25.

Chapter 2 — “Burning Bridges, Drowning Phones”

Mechanic: a throwing mini-game. Christopher phones his boss and **quits**, then throws the phone into **Mary Sue's fish tank**, then **throws the tank itself** (a big physics moment: water spray, fish tumbling in a slow arc, glass shattering). Provide a throw-power meter and a target (the open window or front yard), with fun failures. Mary Sue's dismay is a small reactive moment. Then load the family into the **1994 Ford Taurus**: Frank, Mary Sue and Elsa Ray crammed into the back, and **the letter takes the passenger seat**, buckled in with a seatbelt (a recurring gag). Pan out as the house shrinks.

Chapter 3 — “The Road South” (driving game)

Mechanic: a stylized night driver in 3D, third-person behind the Taurus. Steer, accelerate, brake. Ruts, hills and jumps (“**Up the hill, down the hill, gun it and brake, stay in the air and fall down and shake**”), with a visible **score counter** and a rhythm to it: the “game” takes Christopher’s mind off everything (on-screen: *“just the game and the road and the score and my foot on the pedals”*). **Details:**

- **Black night** “like a racoon chasing the rubbage into the woods.” Racoons dart across the road (dodge them or lose points) and trashcans tumble.
- **The woods on the edge.** Dark trees crowd the roadside with eye-shine. Signage rhymes. Floating text: *“Come this far, no further. Do not leave the door ajar. Stay to the left.”* Hugging the left lane gives a bonus; drifting toward the trees triggers a root-grab hazard.
- **The Inn:** a glowing roadside inn with a sign (*“Do not stay in the inn for the next of kin”*). If the player pulls in, it is a gag failure (bedbug sheets, black coffee) and a quick loop back to the road. **The right choice is to drive on.**
- **Elsa Ray’s voice:** her muffled, buzzing voice layers in as the player drives. A few clear fragments surface, such as *“I won’t look, I won’t speak, but I hear just barely.”* An optional hold-to-listen button slows the game and reveals what she is saying (a small tender payoff that lowers the Flea Meter).
- **Distance:** a HUD odometer that rolls absurdly past thousands (*“the distance I’d driven I couldn’t quite say”*).
- **Arrival:** the sky turns from black to dusty pink; the snow drops away; cacti and heat shimmer appear. Transition card: *5:00 AM. San Antonio.* “It was much more hot and there was no more snow.”

Chapter 4 — “The House in the Clearing” (exploration)

Environment: a **grassy clearing surrounded by thick forest**. A **winding dirt path** leads to the house. The mansion is **oak and red brick, with many, many windows and chimneys galore**. Warm dawn light, mist in the grass, crickets, distant grackles. Make this a stunning establishing shot: a slow aerial approach, then hand control back at the car. **Goal:** go up the steps (“I bounded up the steps with a great force of might”) and enter. Inside is dark. **Locked-door maze:** a hallway of locked doors with *exactly one open door per hallway*, leading to the next. The doors rattle, with tiny gold plaques that rhyme. After the hallways: a **staircase** to the dome room.

Chapter 5 — “The Moon With an Eye” (dome room, dialogue set piece)

Environment: a circular room under a **giant glass dome** with the moon enormous, milky and bright, larger than physically plausible, with a soft rim glow and slow-drifting clouds.

Beats: the moon's **eye** (green and blue) opens. Dialogue, verbatim from the story:

- *“Christopher, old friend, welcome home.”*
- *“Who speaks to me there! No friend of mine!”* (Christopher)
- The Errante: *“Too soon do you forget Christopher... It's been thirty five years since we last spoke, to the day.”*
- Christopher rejects him (*“Nonsense, you glowing moon with an eye...”*).
- The Errante: *“I'm the Errante, your once closest friend... I am the Gronfiser, your uncle is dead, I am your uncle, your father, and friend.”*
- *“You lie again! ... Mention the Gronfiser, or my father, once more, and you'll pay!”*
- The eye vanishes. *“Depart from me now... I will see you when your mind is at ease.”*

Mechanic: dialogue choices that are really tone choices, each affecting the eye's pupil and the Flea Meter. A simple stare-down mini-game: keep your reticle on the eye while the moon's brightness pulses; looking away raises the Flea Meter.

Chapter 6 — “The First Night of Rooms” (darkness, birds, chase)

1. **Pitch darkness.** Only faint moonlight. The player stumbles with a bobbing view. **A vase shatters** (clatter SFX; you can't see it). A “fumble” system: bumping fixtures causes camera shakes.
2. Find the **light switch**. The **great lantern** illuminates: **twelve arms, each holding a bird nest with different birds**, gathering sticks and worms from piles on a **dirt floor with rugs over it**. Build twelve distinct nests (twigs, feathers, eggs) and twelve species by color and silhouette.
3. Mary Sue is at your feet on a rug. Pick her up. The lantern flickers. The birds: *“Leave the girl.”* Christopher: *“Make me.”*
4. **Chase:** run up the stairs, **turn left, right, left some more** (flickering arrows or rhyming hint signs). The birds swarm behind you (a flock of about 60 instanced birds).
5. **The Closet.** Christopher **throws Mary Sue into the closet** (comedic and a bit cruel; the game does not shy away, and her reaction drains a Household Heart). The player then wades through silk, linen and moths.
6. **The birds' chant:** *“The girl, the girl, she must be found, like a kite that must come back to the ground...”*

7. **The Gust.** A great “glutenous” gust of wind sweeps the closet and sucks in the birds, Mary Sue, a **kite** and Christopher. A tumbling cyclone mini-game: dodge flying objects (hangers, feathers, kites) for about 15 seconds with a rotating camera. It stops. **Mary Sue holds the kite; the birds flutter around it.**

Chapter 7 — “Thirteen Candles” (the kitchen)

The Maroon Hallway. Mary Sue says “*Follow me, father.*” She drags the kite and birds down a hallway turned **ruddy ruby maroon by sparse chandeliers**. Many identical doors; she stops at one. **The Kitchen.** Recreate the Wisconsin kitchen with **stained cupboards and tacky marble flooring** (the room Christopher “presided over as emperor”). At the table sit **Frank, Elsa Ray and Mary Sue (at the head)**. Christopher *commandeers* his seat. Elsa Ray gives Mary Sue a **gold-speckled cone hat**; a **pink and white cake with thirteen candles** sits before her. **Beats:**

- **Cake-cutting mini-game:** slice evenly: **one for Frank, one for Elsa, one for each bird (served on the floating kite)**, and finally one for Christopher. Mary Sue: “*Thank you, mother.*” (the odd slip, canon).
- **The Song (rhythm game):** “*Now you must sing!*” Notes scroll toward a hit zone; the birds form a choir. Success makes the camera swirl and the room swell. **Do not use any real birthday-song lyrics**; use an original tune and original rhyming lines.
- **The Transformation.** During the song, slowly dissolve Mary Sue’s features into her *mother’s*: hair, face, creases around the smile, no longer a child. Cut to **Elsa Ray**: hair duller and darker, face thinner and longer. The melancholy beat: Christopher realizes he no longer loves the woman he sees. Stay with this uncomfortable moment; no cheap humor.
- **Frank and Clement.** Frank stares back with **Clement’s sternness, stillness and slyness of smile**. Dialogue: “*You see him now, don’t you.*” / “*Yes I do.*” / “*Then you are ready to see him once more.*” / “*Your grandfather is dead.*” / “*You saw him in me. It is time to see him again.*” He hugs his sister tightly and leaves; the player follows.
- **Corridor to forest.** As you follow Frank, the walls grow **textured and round**, the floor **coarser**, the chandeliers **milky**, until the hallway becomes a forest path. Stars in clusters hang like fixtures.

Chapter 8 — “The Lake” (fishing, memory, grief)

Environment: a clearing opens onto a **glassy lake that never meets the sky**. Texas dusk, violet and teal, stars in clusters, a few fireflies and a **grackle**. **Clement** sits in a **canoe tucked under a tree**, rod cast, a **bobber that never sways**. The moon’s reflection pools

light on the water. **Beats:**

1. Frank: *"There is your father. Go speak to him now. For as long as the Errante will choose to allow."* The father-argument rhyme exchange follows (verbatim). Frank tells him these are *lessons he designed too, before you had me and before you were you*. Frank leaves the edge of the woods.
2. **Christopher boards the canoe** (a wobbly balance step) and speaks to his father. The monologue is a choice-driven rhyme builder where the player picks lines that are warm or boastful (naming his daughter after Clement's mother, naming his son Frank, "the great family we'll be"). **Clement never answers** and never looks down.
3. **The Memory Flash:** the moon's reflection on his father's skin morphs into a **spring evening**. Edith throws Christopher and his sister into the back of her truck, **sobbing**, and guns it down the dirt lane. Child-Christopher's hands stretch out the back window, calling to his father. The father watches from the yard, then turns his eyes up to the moon. Make this a short first-person playable memory with limited movement.
4. **Return:** *"Father, I beg you, speak to me now! This is our last chance 'fore the curtain will bow!"* The bobber drifts. *"God damn you, father. Return to the sky. What good is a life if you watch it pass by."* Christopher turns to find Frank, who is gone, and curses him too ("How does the apple fall so far from the tree?"). The Flea Meter spikes.
5. **The grackle's cry** startles him, and the bobber **tugs** harder. **Fishing mini-game:** a tension-bar reel where patience is rewarded (do not jerk; hold gently). Clement's hands guide the rod as a ghostly overlay. A great **bass** flops into the boat. Short flashbacks: Friday nights loading the truck, choosing lures, going to bed giddy thinking of fish.
6. **Clement's dying wish** surfaces on screen: *take your own children fishing*. Christopher notes he never has. Patience and gentleness. He takes **a deep breath**, looks at the trees, the lake and the reeds. The camera lifts to the **moon**.
7. Fade to white on Clement's voice: **"Hello, son."**

Chapter 9 — "Morning at the Gronfiser's" (eggs, paintings, verse)

1. **Wake** in the enormous bed to chatter through the floorboards and the smell of eggs and buttered toast. **Seven pillows with the Gronfiser's seal:** the player straightens them into a neat row (a small tidy-up mechanic).
2. **Hallway of paintings** in midday sun. Interactive paintings: **the Gronfiser with the pope, who is licking his golden boot; the Gronfiser performing with the Beatles, his signature trombone glittering, entrancing a crowd of 300,000.** Add 3–5 more absurd

original paintings in the same vein (the lion feast, the cannon launch, the elephant ride). Duck the large chandeliers on the stairs (a crouch prompt).

3. **Dining room: a twenty-seat table;** the family sits at the far end. The head seat is open and Christopher takes it, **about eighteen seats from Frank**. The meal is a conversation game: ask each family member about their first night. Frank: *"My bed is big..."* Mary Sue: *"Terrible. I miss all my friends... you threw the phone in the fish tank."* Elsa Ray: *"Whose house was this before, and how on earth did we get it?"*
4. **The Gronfiser Verse.** Christopher recites (rhyming, an animated storybook sequence of 20–30 seconds): gold suits, gimongous gold boots, the gold elephant-head staff with tusks, **George Gervin as his "bestest concubine"** (keep this canon line as written, staged as a comic cameo with a silhouette only), the feasts of **sautéed lion and mushrooms**, the elephant ride, gold coins thrown, the **launch cannon**, the shot into the sky, the eagles, the landing in the splits. The whole state wished he would rule them.
5. **The Hard Question.** Elsa Ray asks why he left this to them, why he clearly loved Christopher so. Christopher's answer verse covers no children of his own, the heir, "by nineteen I was his student." Mary Sue: *"What happened?"* Christopher: *"I left... when they both died."* The room goes silent. Frank: *"Who died, daddy?"* Christopher deflects: *"I met your mother in Milwaukee..."* Elsa Ray gives the half-second glare, then smiles brightly. She sends the kids to the dishes and tells Christopher he must have plenty to do.

Chapter 10 — "The Inventory" (exploration and obsession)

Mechanic: Christopher grabs a **notepad and pen** (the Treasure Notepad), stung that his role as breadwinner was usurped by his uncle. Explore the daytime mansion, now "less restrictive and more forgiving to wandering," and tally items. Required rooms:

- **The Military Gallery** (entered from the kitchen doorway): rows of suits of armor and family crests; each armor holds a **sword, mace, bayonet or gun**. Check each and add it to the notepad. Armor can be poked (it clanks and a shield falls).
- **The Bird Room:** silver, gold and bronze bird statues on the **windowsill, the loveseat, hanging from each blade of the ceiling fan** and protruding from the walls.
- **The Statue Room (red carpet, black walls):** gold statues of the Gronfiser (elephant), **Crumpis** (baboon), **Crumpalump** (tiger), and **Curt**, the first Gronfiser, only **3 feet** tall, with a **loon**. The others stand **8 feet tall**. **The Flea Takeover:** in the statue room, a *funny feeling* overtakes him: stomach churning, heart "quite bold." The screen fills with fleas,

the vignette closes, and a devilish plan forms: **melt it all and sell it, richer than the tsars**. The Flea Meter jumps to about 85 and the visuals go fully manic (gold motes, heat haze, a rhyming HUD).

Chapter 11 — “The Melting Spree” (final set piece)

A multi-stage sandbox finale. Keep each stage playable and funny.

1. **Rally the Family:** run to the kitchen and scream the rhyme. **The family grabs garbage bags** and loots silverware, candlesticks and trinkets “smaller than a mouse.” A timed collection mini-game: fill the bags before time runs out. The family carries the loot onto the lawn.
2. **Elsa Ray’s Confrontation.** She lingers: *“Christopher, I worry about you... you ignored [Mary Sue’s] birthday which is true. Now you’re selling off your uncle’s gold, I worry that your heart’s gone cold.”* Dialogue choices (all rhyming). She reminds him *“remember who moved south with you.”* He kisses her lightly and **bolts into the woods** (the player may choose to stay, which lowers the meter, but Christopher bolts anyway: *“Just give me a little time, I promise you that I’ll be fine”*).
3. **The Axe:** find the **axe behind the shed**. A **chop-timing mini-game** in the rhythm of *chop chop chop, hack hack hack*: fell trees, **stack them, split logs** on a stump. Between sets, **whisky sours**: drinking refills stamina, but each drink blurs the screen and wobbles the controls. **Several hours pass** (day-to-dusk lighting).
4. **The East Wing:** the fleas whisper *go, you must go visit the east wing*. Navigate **cobwebs and torn tapestries**; clear the gunk with your hands. Find the **gimongous pot**, decorated with painted elephants.
5. **Carry the Pot:** a comic physics challenge. Balance it on his **neck**, then **tumble down the stairs** (ragdoll fall, face hits the deck). Pick it up, take it out the door and **roll it down the slope** to the woodpile.
6. **Light the Fire:** matches, ablaze (“it burned so bright I averted my gaze”). Throw the pot onto the logs. Back to the house for a **whiskey shot**.
7. **Feed the Fire:** throw in the metal: **five chandeliers, nine dining sets and a necklace**, all told. A counter keeps tally in the notepad. Let the player over-feed for fun, but cap progression at the canon items.
8. **The Dance:** Christopher **lathers himself in oil**, finds a **grand flute imported from France**, and dances in rings around the fire while playing it. A flute-by-keys mini-game that distorts as the heat climbs; the music “dements.” After a while he sings (an original rhyming song; the player picks the rhymes).

9. **Collapse.** He checks that the fire will burn through the night, staggers into the house in his heat-dazed drunken state, climbs the stairs, walks the hallway and **collapses in bed beside sleeping Elsa Ray.** Final shot: the camera drifts from the bed up to the window, where the **moon hangs outside and its eye slowly opens.** Cut to black.

Credits (optional creative flourish): a title card, *"I've Hung You From the Moon,"* with tiny silhouettes of the family dangling from the moon's crescent like ornaments while the Errante's eye winks.

8. Mansion Map (keep spatially consistent)

Ground floor: foyer; dining room (20-seat table); kitchen (a doorway into the military gallery); military gallery; bird statue room; red-carpet statue room. Upper floor: bedrooms, hallways of paintings, a closet, the maroon corridor. Top floor: the dome room. Wing: the east wing (cobwebs, the smelting pot). Outside: the dirt path, front steps, grassy clearing, shed with the axe, woodpile slope and thick forest. In dream-logic chapters (6, 7, 8) rooms may defy this map, but the *real* daytime house (Chapters 9–11) must stay stable and navigable.

9. UI and HUD

Minimal and storybook-styled. Top-left: the Flea jar. Top-right: the Household Heart (four small hearts). Bottom: the caption box. Each chapter opens with a title card in gold type on a parchment texture. Pause menu: resume, quality toggle, volume, chapter select, controls. Loading screen: an animated moon with a blinking eye.

10. Canon Accuracy Checklist (do not violate)

- The story begins **September 14**, during the annual rerun of *The Sound of Music*.
- The TV is a **27-inch RCA CRT**, hit twice with a **broom handle**.
- **The Sweaty Man:** bathrobe, barefoot, long scraggly beard, a long finger to the lips, a damp letter *"baking like the sun."*
- The letter is from **Joshua Primo**, **"the Gronfiser,"** after thirteen years of unanswered letters; Wisconsin to San Antonio.
- The family: **Christopher, Elsa Ray, Mary Sue (turns 13), Frank.** The story's one stray "John" at the kitchen table is a typo; use **Frank**.

- **1994 Ford Taurus**; the letter rides in the passenger seat. Phone and fish tank thrown (**the fish tank is Mary Sue's**).
- Arrival at **5 AM**; **oak and red brick** house; grassy clearing; thick forest; many windows and chimneys.
- The Errante's eye is **green and blue**. "Thirty five years since we last spoke."
- **Twelve** bird-nest arms; **seven** pillows; **twenty-seat** table; sitting **eighteen** seats from Frank; **thirteen** candles.
- Gold statues: Joshua Primo (elephant), **Crumpis** (baboon), **Crumpalump** (tiger), **Curt** (loon, 3 ft). The others are 8 ft.
- Paintings: the pope licking a golden boot; the Beatles with a trombone and a crowd of **300,000**.
- Clement (father), **Edith** (mother), and a sister exist; the father is dead; his dying wish was about taking your own children fishing.
- Melted: **five chandeliers, nine dining sets, a necklace**. The pot is decorated with **painted elephants**. The flute is **French**.
- Ending: Christopher collapses into bed beside Elsa Ray.

11. Build Order and Acceptance Criteria

1. Core engine, input, state, UI, audio bus, post-processing, scene manager and the debug chapter select.
2. The moon asset (with its eye) and the Dialogue system; build the title screen with the moon blinking as the hero shot.
3. Chapters 1 through 3 as a complete vertical slice.
4. Chapters 4 through 7, then 8 through 11.
5. A polish pass: lighting, particles, sound, text timing, meter feedback and performance.

Acceptance: a first-time player can complete the game from start to finish with no console errors; every chapter has at least one distinct playable mechanic; every canon detail in Section 10 is visible in-game; the Flea Meter visibly changes the look and sound; and the final shot of the moon opening its eye lands emotionally. Make it strange, funny, handsome and a little heartbreaking.

12. Merged Direction (Addendum: wins over earlier sections wherever they conflict)

This addendum folds in the strongest ideas from a second creative brief and reconciles them with what a browser can actually run smoothly. Sections 0–11 remain the chapter-by-chapter spec and the canon authority.

12.1 The governing principle: every place is a piece of Christopher's interior

The mansion is not just a haunted house, the moon is not just a monster, the birds are not just enemies and the lake is not just a location. Design and light each environment so it says something about him:

- **Wisconsin living room:** control as domesticity. He rules a small kingdom, and the TV he destroys is the family's shared attention.
- **The road:** escape as numbness. The driving "game" replaces feeling with a score.
- **The mansion at night:** inherited worth he cannot navigate. Locked doors, one open path.
- **The dome:** being seen. Intimacy before menace.
- **The bird lantern hall:** a nurturing instinct (the birds guard Mary Sue) that he treats as a threat to his ownership.
- **The birthday kitchen:** time he missed. The room is a memory he cannot trust.
- **The lake:** the silence he inherited. Spacious and emotionally exposed.
- **The daytime mansion:** value over people.
- **The smelting fire:** compulsion. Beauty with an unstable edge.

12.2 Baseline, then violation

Spend the first 20–40 seconds of each location making it **physically believable**, then break exactly one rule: room dimensions, gravity, sound, or who is sitting at the table. The break only lands if the world was convincing beforehand. Example: the kitchen is a perfect Wisconsin kitchen with correct cupboards and hum of a fridge, and only then do the chandeliers go milky and the walls turn round.

12.3 Ambiguity rules (do not resolve these)

- **The Errante** claims to be Christopher's once-closest friend, and also his uncle, his father and the Gronfiser. Never confirm or deny. No cutscene, label or codex entry may say who or what he is.

- **Clement** never speaks until the final “Hello, son.” Do not explain whether the lake is a memory, a gift or a lesson.
- **Frank** is never explained as a ghost, a vision or a real boy. His dream-chapter calm is simply delivered.
- “**When they both died**” is never clarified. **Edith’s flight** is never explained.
- Where the text is surreal or contradictory, let it stay that way. Add no lore the story does not contain.

12.4 Tone rules

- No generic horror: no jump scares, no gore, no conventional spooky dialogue. A gold statue can be funny, the eye can be frightening, the cake can be warm and the silent father can be devastating. All of these coexist.
- **Lake silence:** no music for the first 60 seconds of the lake; only water, crickets, a grackle and breath.
- **The moon’s eye should read as a presence, not a sticker.** Build it as part of the moon: a recessed crater socket, lid-like crescent ridges formed from the lunar surface, a wet cornea that reflects the dome, a radial-fiber green-and-blue iris with slow micro-movements, and faint green-blue light cast onto the room whenever it is open.

12.5 New mechanic: the Appraisal Gaze

From Chapter 9 onward Christopher’s attention shifts from people to objects, and the game shows it:

- Objects sprout small gold price tags and glints; family members become slightly desaturated and softer-edged as the Flea Meter rises.
- By Chapter 11 the family is rendered as flat, dim silhouettes unless the player **holds F to “see them,”** which restores color, lowers the Flea Meter and shows a tender micro-moment (Frank’s sleepy wave, Mary Sue with the cone hat still on).
- The player can ignore this entirely and still finish. It colors the epilogue lines only.

12.6 Color script refinements

Use the per-chapter palettes in Section 4 plus these rules: keep **local material colors** (brick stays brick); apply chapter grading as light and mild fog tint, **never a uniform filter**; forest transition is deep green, dusky blue and muted silver; the daytime mansion is gold, polished wood, red carpet and white daylight; the smelting area is orange firelight, blackened wood and molten gold with heat-haze distortion on the trees.

12.7 Web-performance reconciliation: “stylized realism”

The goal is a world that *feels* weighty and detailed while staying fast. **Replace any ambition for fully physical lighting, real volumetrics or heavy PBR with these techniques:**

- **Fake the expensive things.** Baked-looking lighting: canvas-painted lightmaps and ambient-occlusion gradients in textures, vertex-color shading in corners, contact-shadow decal blobs under objects. Volumetric light is additive cone meshes and billboarded sprites. Heat haze is a screen-space UV wobble in the final pass. Reflections use one small shared cubemap, rendered once.
- **Material tiers.** `MeshStandardMaterial` only on hero props (gold, glass, the moon, the Taurus). Everything else uses `MeshLambertMaterial` or `MeshBasicMaterial` with baked-looking textures.
- **Budgets.** At most about 300 draw calls and about 300k visible triangles per scene; textures 512–1024 px and shared via atlases where possible; **one** shadow-casting light per scene (two at most); instancing for anything repeated more than 10 times.
- **Scene streaming.** Build one chapter at a time, dispose of the previous one fully, and show an animated moon loading card during `build()`. Never keep two chapters resident except during a short transition.
- **Adaptive quality.** Measure frame time; if it averages above 20 ms, lower `renderer.setPixelRatio`, then disable post-processing, then drop shadows. A visible Quality setting (Auto / High / Low) lets the player override.
- **Single post pass.** One full-screen shader combining vignette, grain, chromatic aberration, color grade and a cheap bloom approximation, optionally at reduced resolution.
- **Housekeeping.** Pause the render loop when the tab is hidden; start audio only after the first user gesture; no external assets; target a file under about 1.5 MB and a first interactive frame in under 3 seconds; wrap each scene build in try/catch with a graceful fallback message rather than a white screen.

12.8 The detail rule: no hiding primitives in the dark

Fog, darkness and bloom may not be used to conceal crude models. Every hero prop must hold up when the player walks up to it. Use a **procedural detail kit** to build real silhouettes:

- `LatheGeometry`: vases, candlesticks, pots, lamp bases, bobbers, the flute, bird bodies.
- `ExtrudeGeometry` with bevel: picture frames, armor plates, the TV bezel, door panels, moldings.

- **TubeGeometry** : chandelier arms, twig nests, fishing line, vines, rope.
- **InstancedMesh** : bricks and mortar accents, books, candles, leaves, feathers, crystals.
- Canvas-painted texture details for wear, stains, grain, wallpaper pattern and scuffs.

Minimum hero-prop detail targets:

- **Chandeliers**: at least 8 arms, a central column and hanging crystal pendants, with a distinct style per chapter (warm brass, ruby, milky).
- **The bird lantern**: twelve arms, each with a woven twig nest, eggs or chicks and straw detail.
- **Armor**: at least 12 distinct pieces (helm, gorget, pauldrons, cuirass, gauntlets, greaves) plus a different weapon each.
- **The Taurus**: beveled body, wheel wells, headlight cones, a rear window with the family's silhouettes, and the buckled letter in the passenger seat.
- **The RCA TV**: a convex screen with scanlines, a wooden-effect case, knobs, a power glow and a cracked-glass break effect.
- **The mansion exterior**: textured brick with window surrounds, a roofline with at least 8 chimneys, porch columns and shuttered windows.
- **The gold statues**: stylized but proportioned, each with its animal staff carved distinctly.
- **The pot**: a huge lathe-turned form with painted elephants in a frieze, rivets and heat-darkened rim.

12.9 Source ending (clarified)

The story does not stop at the start of the melting plan. It runs through the chopping, the pot, the fire, the flute and the dance, then Christopher collapsing in bed beside Elsa Ray. Section 7's ending stands. The final shot of the moon opening its eye is an optional flourish, held ambiguous and unexplained.

12.10 Delivery instructions for the builder

- **Actually build it.** Do not stop at a design note or a title screen. If you have a coding environment, create the file, run it, test it and fix errors. If you cannot run code, write conservatively and re-read your own code for errors before finishing.
- **Technology**: one `.html` file, plain JavaScript (no TypeScript, no build step, no ES module imports, since `file://` loading breaks them), Three.js from cdnjs only.

- **Test hooks:** support `?chapter=N&debug=1` to jump to a chapter, a visible FPS and draw-call readout in debug mode, and a **self-test mode** that loads every chapter in turn, runs it for a couple of seconds, and logs any errors and frame times to the console.
- **If you run out of budget, cut in this order:** (Tier 3) credits flourish, touch controls, extra paintings, the Appraisal Gaze polish; (Tier 2) the flute mini-game, whisky blur, chop-timing depth, closet cyclone depth; (Tier 1, never cut) the canon beats of every chapter, the moon with its eye, the Flea Meter, the letter, TV and broom sequence, the drive, the dome dialogue, the birthday transformation, the lake with “Hello, son,” the Gronfiser verse, the statue-room flea takeover and the fire.
- **Finish with an honest report** of what is fully implemented, what is simplified and what was cut.

PART II — IMPLEMENTATION SPECIFICATION

13. How Part II Works

Part I (Sections 0–12) says **what** to build and **why**. Part II (Sections 13–22) says **how**, with concrete numbers, data shapes, state machines and test steps. **If a number or rule in Part II conflicts with Part I, Part II wins.** Every number here is a starting value: put them all in one `CONFIG` object at the top of the file so they can be tuned in one place.

Conventions: units are meters, seconds and radians. +Y is up. Each scene’s player spawn faces –Z. Colors are written as hex. When the spec says “flag,” it means a boolean in `State.flags`. When it says “delta,” it means an additive change to a meter. Names in code `font` are suggested identifiers; the builder may rename them consistently.

14. Code Architecture

14.1 Region order inside the single HTML file

Use banner comments so the file stays navigable. In this order:

1. `CONFIG` (all tunables), `UTIL` (seeded RNG `mulberry32`, clamp, lerp, smoothstep, easing functions `inOutCubic`, `outBack`, `outElastic`, vector helpers).
2. `TEX`: the canvas texture factory (Section 16.6).

3. **GEO** : the procedural detail kit (lathe, extrude, tube, instancing helpers) and prop builders such as `buildChandelier(opts)` , `buildArmor(opts)` , `buildNest(opts)` , `buildTV()` , `buildTaurus()` .
4. **MAT** : a material cache keyed by name, so identical materials are shared.
5. **FX** : post-processing pass, particle systems, light-shaft cones, screen-space flea overlay.
6. **AUDIO** : engine, music sequencer, SFX recipes, voice blips.
7. **INPUT** : keyboard, mouse, pointer lock, touch, action mapping.
8. **UI** : DOM overlay builders (captions, HUD, menus, letter, notepad).
9. **DIALOGUE** and **CUTSCENE** : runners.
10. **STATE** : the state object, event bus, objectives, hints.
11. **PHYSICS** : collision, triggers, floors, simple rigid bodies for thrown objects.
12. **SCENES** : base `Scene` class, then `TitleScene` , `Ch01` ... `Ch11` , `Epilogue` .
13. **MAIN** : boot, loading, resize, loop, error handling, debug tools.

14.2 The game loop

- Use `requestAnimationFrame` . Keep an accumulator and step logic at a fixed `1/60` s, with the frame delta clamped to `0.1` s so tab-switching does not cause huge jumps.
- Per logic step, in order: `Input.poll()` → `Scene.update(dt)` → `PHYSICS.step(dt)` → `Dialogue/Cutscene.update(dt)` → `Audio.update(dt)` → `UI.update(dt)` .
- Per render: scene render to a render target (or direct if post is off), then the post pass, then DOM updates that need the camera (billboard labels).
- A global `GameTime` provides `wait(seconds)` that respects pause, so cutscene coroutines freeze when the game is paused.
- Pause on `Esc` , on pointer-lock loss and on `visibilitychange` (hidden).

14.3 State object and event bus

State shape (suggested):

- `chapter` (integer 0-12), `time` (total seconds), `flea` (0-100), `hearts` (`{elsa, marySue, frank, self}` , each 0-10, starting at 6), `flags` (object of booleans), `counters` (object of numbers, e.g. `kicks` , `fishCaught` , `treasureValue`), `notepad` (array of `{id, name, value, room}`), `inventory` (array of ids), `settings` (`quality` , `volume` , `sensitivity` , `reducedMotion` , `subtitles` , `invertY`).

The event bus has `Bus.on(name, fn)`, `Bus.off(name, fn)` and `Bus.emit(name, payload)`. Required event names: `interact`, `flea`, `heart`, `flag`, `dialogue:start`, `dialogue:end`, `cutscene:start`, `cutscene:end`, `objective:set`, `objective:done`, `chapter:complete`, `quality:change`.

Helpers: `addFlea(n, reason)` clamps to 0-100 and emits `flea`; `addHeart(who, n)`; `setFlag(name)`; `hasFlag(name)`. Every meter change should spawn a tiny floating HUD tick so the player learns the cause and effect.

14.4 Interactable schema

Every interactable is registered with `Interact.add({ id, object, prompt, range, enabled, once, onUse, highlight })`:

- `object`: a Three.js `Object3D` whose bounds define the hit volume (or an explicit invisible hit box).
- `prompt`: the label, e.g. "Open letter," "Kick," "Examine bird."
- `range`: default 2.4 m.
- `enabled`: a function returning boolean (gated by flags).
- `once`: disable after use.
- `onUse(ctx)`: runs the action; `ctx` carries `scene`, `state` and helpers.
- `highlight`: draw an outline when targeted (a back-face scaled clone shell with an unlit emissive material, enabled only for the current target to save draw calls).

Targeting: each logic step, cast a ray from the camera center, test registered interactables within range, choose the nearest in front, and show the prompt `[E] <prompt>`. Also show a small reticle dot that grows when something is targetable.

14.5 Dialogue script format

Author all dialogue as data. Example shape:

- `Dialogue.run({ id:'moon_01', start:'a', nodes:{ a:{ who:'moon', text:'Christopher, old friend, welcome home.', next:'b' }, b:{ who:'chris', text:'Who speaks to me there! No friend of mine!', next:'c' }, c:{ who:'moon', text:'...', choices:[{ text:'...', goto:'d', flea:+3, heart:{ self:-1 }, set:'rudeToMoon' }, { text:'...', goto:'d', flea:-2, canon:true }] } } })`

- A node may carry `cam` (a camera preset to ease to), `sfx`, `emote` (a character animation name), `wait` (seconds), and `onEnter` / `onExit` callbacks.
- Typewriter speed: 38 characters per second; pause 180 ms after `.`, `?`, `!`; 90 ms after `,`; click or `Space` completes the line, and a second press advances.
- Choices show as 2–4 buttons selectable by mouse or keys `1–4`. Exactly one choice per fork is marked `canon:true`; all others must `goto` back into the canon path within one or two nodes.
- A speaker's name appears in a small tag above the caption box in that character's color. The caption box has a "log" button that reopens the last 12 lines.

14.6 Cutscene runner

Cutscenes are `async` functions using a small API that respects `GameTime`:

- `await Cut.cam({ pos:[x,y,z], look:[x,y,z], fov, dur, ease })`
- `await Cut.wait(s)`
- `await Cut.say(who, text)` (uses the dialogue UI, auto-advance optional)
- `await Cut.anim(actor, name)` and `Cut.play(sfxName)`
- `Cut.fade(toBlack|toWhite|clear, dur)`
- `Cut.letterbox(true|false)`
- `Cut.lockPlayer(true|false)` Every cutscene is skippable with a hold of `Space` (a small ring fills over 1 second), and skipping must jump all state to its end values (flags, positions, meters) so the next beat is consistent.

14.7 Objectives and the hint system

- One active objective at a time, shown in the top-center as a short rhymed line (e.g., *"Find the switch before the dark finds you."*). `Objectives.set(id, text, targetObject?)`.
- **Hint ladder:** if the objective is not completed after 45 s, show a rhymed hint line; after 90 s, pulse a soft glow on the target object; after 150 s, show a faint compass arrow on screen. Reset the timers whenever the player completes a sub-step.
- Every objective must have `hint1`, `hint2` strings, and a `targetObject` where one exists.

14.8 Scene contract

Each scene class implements `build()` (create all meshes and register colliders), `enter()` (position the player, start music and ambience, fire the opening cutscene), `update(dt)`, `exit()` and `dispose()` (traverse and dispose geometries, materials, textures and render targets). Base-class helpers: `addCollider(box)`, `addTrigger(box, onEnter, onExit)`, `addFloor(rect)`, `addInteract(def)`, `addLight(light)`, `spawnPlayer(pos, yaw)`, `complete()` (emits `chapter:complete` and begins the transition to the next scene). Transitions: fade to black over 0.8 s, dispose the old scene, build the new one while the moon loading card shows, fade in over 0.8 s. If a scene's `build()` throws, catch it, show a friendly error with a “Skip to next chapter” button, and log the stack to the console.

14.9 Three.js r128 pitfalls (the pinned version)

- Use `renderer.outputEncoding = THREE.sRGBEncoding` (not `outputColorSpace`), and set `texture.encoding = THREE.sRGBEncoding` on color textures drawn from canvases. Leave data textures (bump, roughness) as linear.
- `THREE.Geometry` no longer exists; use `BufferGeometry`. `CapsuleGeometry` is not available (use a cylinder plus two spheres). There is no `TextGeometry`, `Sky`, `Reflector` or `Water` in the core build; use DOM text, canvas sprites and your own shaders.
- Use `THREE.MathUtils` (not `THREE.Math`). Use `new THREE.Euler(0, 0, 0, 'YXZ')` for FPS look.
- `InstancedMesh` with per-instance color needs `instanceColor`; call `instanceMatrix.needsUpdate = true` after changes.
- Shadow lights: set `light.shadow.bias = -0.0005` and `light.shadow.normalBias = 0.02` to avoid acne; set `castShadow` only on meshes that need it.
- `Points` materials need `depthWrite:false`, `transparent:true`, and `blending: THREE.AdditiveBlending` for glow.
- Browsers require a user gesture before audio and pointer lock: the title screen's “Begin” button must `resume()` the `AudioContext` and request pointer lock.
- Call `renderer.setPixelRatio(Math.min(devicePixelRatio, CONFIG.maxPixelRatio))` (default max 1.5, 1.0 on Low) and resize the post render target when the window changes.

15. Player, Camera and Collision

15.1 Player controller constants

- Eye height `1.65`, crouch eye height `1.05` (toggle with `C`), body radius `0.35`, body height `1.8`.
- Walk speed `3.2` m/s, sprint `5.2` m/s, crouch `1.6` m/s. Acceleration `18` m/s², deceleration `22` m/s² (so movement feels grounded, not floaty).
- Gravity `12` m/s², jump impulse `4.2` m/s (about 0.7 m apex). Step-up height `0.35` m.
- Mouse sensitivity `0.0022` rad/pixel, pitch clamped to $\pm 85^\circ$, smoothing off by default. Base FOV `70`; sprint widens to `76` over 0.2 s.
- Head bob: while walking, vertical `0.035` m at `7.5` Hz and lateral `0.02` m at `3.75` Hz; amplitude $\times 1.4$ when sprinting; disabled when `reducedMotion` is on.
- **Drunk modifier** (used in Chapter 11): adds camera roll $\pm 3^\circ$ at `0.4` Hz, input steering lag `0.15` s and a gentle sway on movement direction, all scaled by `drunk` (0–1).
- **Flea modifier**: at `flea > 60` add a subtle FOV breathe ($\pm 2^\circ$ at `0.25` Hz); at `flea > 80` add periodic micro-jitter in look direction (`0.002` rad at `9` Hz) and slightly increase the head bob.

15.2 Collision model

Keep collision simple and robust:

- **Colliders**: axis-aligned boxes `{x0,x1,y0,y1,z0,z1}` in world space (rotated furniture is approximated by boxes). Each scene registers its colliders in `build()`.
- **Resolution**: move X first, resolve against boxes whose Y range overlaps the player's body; then move Z and resolve. Treat the player as a circle (radius 0.35) in the XZ plane against each box expanded by the radius.
- **Floors**: a list of `{x0,x1,z0,z1,y}` flat floor rectangles and `{x0,x1,z0,z1,y0,y1,axis}` ramps (stairs are modeled visually as steps but collided as a ramp). `floorY(x,z)` returns the highest floor under the player that is within step-up height. If there is no floor, apply gravity (fall) and respawn at the last safe position if `y < -20`.
- **Triggers**: boxes with `onEnter` and `onExit`, used for cutscene starts, room transitions and ambience crossfades.
- **Soft bounds**: every outdoor scene has an invisible circular boundary with a gentle pushback and a rhymed line, so the player never wanders into empty void.

- **Thrown objects** (phone, tank, loot, fish) use a minimal rigid-body helper: position, velocity, angular velocity, gravity, one-bounce restitution `0.35`, floor and box collision, and a lifetime.

15.3 Camera system

Modes: `FPS` (default), `Rail` (a Catmull-Rom spline for flythroughs), `Fixed` (a locked position with a look target), `Chase` (behind the Taurus, springy follow) and `Seat` (locked to a seated viewpoint with limited look range, used for the car-back seat, the canoe and the dining table).

- `Camera.to(preset, dur, ease)` smoothly interpolates position, look target and FOV.
- **Shake:** a trauma value in 0–1 decaying at `1.5 /s`; offset = `trauma2 × smoothed noise` on position (± 0.08 m) and roll ($\pm 2^\circ$). Disabled when `reducedMotion` is on.
- **Letterbox:** two black bars sliding to 11% of screen height each in 0.5 s.
- **Seat mode look range:** yaw $\pm 90^\circ$, pitch $\pm 40^\circ$, so the player can look around while seated but not break the framing.

16. Rendering Specification

16.1 Renderer setup

- `new THREE.WebGLRenderer({ antialias:true, powerPreference:'high-performance', alpha:false }); outputEncoding = sRGBEncoding; toneMapping = THREE.ACESFilmicToneMapping; per-scene toneMappingExposure` from the grade table below.
- `shadowMap.enabled = true, type = THREE.PCFSoftShadowMap` on High, disabled on Low.
- Scene background and fog set per chapter: `THREE.FogExp2` for outdoor and large interiors (density 0.015–0.04), linear `Fog` for small rooms.
- If post-processing is on, render the scene to a `WebGLRenderTarget` (`RGBAFormat`, `HalfFloatType` if supported, otherwise `UnsignedByteType`) with depth buffer, at pixel-ratio scale. On Low, render directly to the canvas and fake the vignette with a CSS radial-gradient overlay.
- Frame-time governor: keep a rolling 60-frame average. Above 20 ms for 2 seconds → step down one level: (1) pixel ratio $\times 0.8$ (floor 0.6), (2) turn off bloom, (3) turn off shadows, (4) turn off post pass. Never step back up automatically more than once per

10 seconds, and only if the average is under 12 ms.

16.2 Post-processing pass (one full-screen shader)

Uniforms: `tScene`, `uTime`, `uVignette`, `uGrain`, `uChroma`, `uExposure`, `uSaturation`, `uTint` (vec3), `uContrast`, `uBloomAmt`, `uFlea` (0-1), `uHeat` (0-1), `uDrunk` (0-1), `uFade` (vec4 color + alpha).

- **Vignette:** smooth radial darkening, `uVignette` 0.2-0.7.
- **Grain:** hash noise $\times$ `uGrain`, animated by time, 0.02-0.07.
- **Chromatic aberration:** sample R and B at $\pm$ `uChroma` radial offset (0-0.006); multiply by $(1 + uFlea \times 2)$.
- **Cheap bloom:** a 5-tap blur of bright pixels (luminance > 0.85) added at `uBloomAmt` (0-0.35); if budget allows, do a half-resolution blur pass.
- **Heat haze:** UV displacement by animated sin-noise, strength `uHeat` $\times$ 0.004, strongest around the pot position in Chapter 11 (pass the pot's screen-space center as `uHeatCenter`).
- **Drunk:** a soft double-image (sample at a small offset $\times$ `uDrunk`) plus a slow sinusoidal wobble.
- **Flea overlay:** a procedural "crawling dots" pattern near the screen edges (small bright specks following noise-driven paths), intensity `uFlea`. If heavy, render the specks as 30-80 absolutely-positioned DOM elements instead.

16.3 Per-chapter grade values (exposure / saturation / tint / vignette / grain / chroma)

- **Title:** 1.0 / 1.0 / #cfd6ff / 0.55 / 0.05 / 0.002
- **Ch1 Wisconsin:** 1.05 / 0.95 / #ffd9a8 / 0.4 / 0.04 / 0.001
- **Ch2 Departure:** 1.0 / 1.0 / #ffe0b8 / 0.45 / 0.04 / 0.002
- **Ch3 Road:** 0.9 / 0.85 / #9db3d9 / 0.6 / 0.06 / 0.003
- **Ch4 Dawn house:** 1.1 / 1.05 / #ffc9a3 / 0.35 / 0.03 / 0.001
- **Ch5 Dome:** 0.95 / 0.9 / #b9d6ff / 0.55 / 0.04 / 0.003
- **Ch6 Dark rooms and lantern hall:** 0.8 then 1.0 once lit / 0.9 / #c9ffb8 / 0.65 / 0.06 / 0.004
- **Ch7 Maroon hall and kitchen:** 1.0 / 1.0 / #ffd2a0 (kitchen), #ff9aa8 (hall) / 0.45 / 0.04 / 0.002
- **Ch8 Lake:** 0.85 / 0.8 / #9fc3d6 / 0.6 / 0.05 / 0.002

- **Ch9 Morning:** 1.15 / 1.05 / #fff1cf / 0.3 / 0.02 / 0.0
- **Ch10 Inventory:** 1.1 / 1.1 / #ffe08a / 0.4 / 0.03 / 0.001 (+flea effects as the meter rises)
- **Ch11 Fire:** 0.9 / 1.1 / #ffb15c / 0.55 / 0.06 / 0.004
- Crossfade grades over 1.5 s inside scenes where the mood shifts (e.g., Ch6 after the switch is flipped). These are **lighting and mild tint only**; never wash local colors out.

16.4 Lighting recipes

- **Hero light:** the one shadow-casting light per scene. Moon scenes: a `DirectionalLight` (color #cfe0ff, intensity 0.9) with a 2048 shadow map covering only the area near the player (follow the player, snap the shadow camera to texels to prevent shimmer). Interiors: a `SpotLight` or `PointLight` where the story motivates it (the lantern, the TV).
- **Practicals:** candles, lamps and chandeliers are emissive meshes plus **fake lights:** a billboard glow sprite (additive) and, for at most 4 nearby, real `PointLight`s with distance 6–10. Flicker: $\text{intensity} \times (0.9 + 0.1 \times \text{noise}(t \times 8))$.
- **Ambient:** `HemisphereLight` (sky, ground, 0.25–0.6). Add a very low `AmbientLight` (0.08) in the darkest scenes so geometry is never fully invisible.
- **Contact shadows:** a soft radial-gradient decal under furniture and characters (a plane with a canvas texture, `depthWrite:false`).
- **Baked-look AO:** darken corners with vertex colors on large room meshes (multiply by corner proximity) or paint an AO gradient into wall textures near floors and ceilings.

16.5 Particles and light shafts

- **Particle system:** a reusable `Emitter` class using `Points` with a ring-buffer; parameters: count, lifetime, size curve, color curve, velocity, spread, gravity, blending. Presets: `dust`, `snow`, `fireflies`, `embers`, `feathers`, `coins`, `fleas`, `smoke`, `waterSpray`, `sparks`, `moths`.
- **Light shafts:** open-ended cone meshes with an additive material and a vertical-gradient alpha texture, opacity 0.06–0.15, slightly rotating dust inside. Use for the dome, windows at midday and the TV beam.
- **God-ray fake:** for the moon and the fire, a large soft billboard sprite behind the object plus a few thin additive streaks.

16.6 Texture factory (all drawn on `<canvas>`)

Each function returns a `CanvasTexture` (with `wrapS/T = RepeatWrapping` , `anisotropy min(4, max)`), plus an optional bump canvas. Resolution 512 unless noted.

- `wood(color, plankCount, grain)` : vertical planks with color variation per plank, noise grain lines, dark seams, scattered knots.
- `brick(color, mortar)` : staggered rows, per-brick color jitter, mortar lines, fine noise; 1024 for the mansion facade.
- `damask(c1, c2)` : a repeating symmetric ornament drawn with bezier curves over a base color.
- `marbleTile(c1, c2)` : a checkerboard (kitchen) with veining via random bezier strokes and slight per-tile tint; used for “tacky marble.”
- `carpet(color)` : noisy weave pattern with a border band; for the red-carpet statue room.
- `plaster(color)` : mottled noise with water-stain gradients near the top.
- `grass()` , `dirt()` , `bark()` , `leaves()` (alpha-masked cards), `water()` (animated via shader), `moon()` (layered fractal noise, craters via radial gradients, maria patches), `iris()` (radial fibers in green and blue with a dark pupil slit and highlights), `cake()` (pink-and-white swirl), `linen()` , `gold()` (subtle streak noise), `parchment()` (aged paper with stains and torn edges), `crt()` (scanline and convex-glow overlay, animated with static).
- Texture memory budget: keep the total under about 40 MB of GPU memory per scene; reuse and tile aggressively.

16.7 The moon and its eye (hero asset)

- **Moon body:** a high-resolution sphere (`SphereGeometry(30, 96, 64)`) with the moon texture, a bump map, and a custom `onBeforeCompile` or a second slightly larger back-face sphere for a Fresnel rim glow. Place it so it fills a large part of the dome view (apparent radius $\approx 18^\circ$).
- **Eye socket:** a circular depression: a radial vertex displacement on the sphere (or a decal ring mesh) with crater-rim lids: two crescent-shaped meshes (upper and lower lid) with moon texture that rotate around the eye center to open and close. Brow: a darker maria patch arching above.

- **Eyeball:** a small sphere recessed in the socket: a glossy cornea shell (transparent, high specular, env reflection of the dome) over an iris disc (the `iris()` texture, emissive 0.6) with a pupil that dilates (scale 0.4–1.2) based on the Errante's mood.
- **Behavior:** blinks every 4–9 s (lids close in 0.12 s and open in 0.2 s); gaze tracks the player with a 0.4 s lag and small micro-saccades every 0.7–1.8 s; opens slowly (1.5 s) at the start of the dialogue; vanishes at the end by closing then fading the iris glow and sliding the lids shut.
- **Light:** a faint green-blue `PointLight` at the eye (intensity $0.4 \times \text{openness}$) lighting the room.
- **Reuse:** the same asset appears in the title screen, Chapter 5, as a distant moon in Chapters 6, 8 and 9 windows, and the final shot (eye open).

16.8 Characters: construction and animation

- **Rig:** nested `Group` s (hips → spine → chest → head; shoulders → upper arm → forearm → hand; hips → thigh → shin → foot). Bodies from tapered cylinders and spheres; heads from a sphere with a canvas face texture (eyes, brows, mouth shapes swapped by changing texture offset on a small atlas of 6 expressions: neutral, smile, frown, shock, shout, sad).
- **Character styles:** Christopher (broad, rumpled, five-o'clock shadow), Elsa Ray (slender, blue eyes, hair mesh from a stretched hemisphere), Mary Sue (small, gold hair, cone hat prop), Frank (small, stiff posture), the Sweaty Man (tall, thin, bathrobe cloth plane with a sine wave in vertex shader, steam emitter), Clement (seated, still, hat), Edith (silhouette only), the birds (see 16.9), statues (stylized gold figures, `metalness 1`).
- **Animation set:** `idle`, `walk`, `run`, `sit`, `eat`, `sing` (mouth cycling), `clap`, `wave`, `point`, `hug`, `throw`, `kick`, `stumble`, `faint`, `reel`. Each is a small procedural function of time that sets joint rotations. No skeletal assets.
- **Expression system:** `setExpression(actor, name)` plus a talking mouth flap synced to the typewriter characters.

16.9 Birds

- Two LOD levels: a **hero bird** (lathe body, cone beak, two wing planes with 3 feather layers, tail fan) for close-ups, and an **instanced flock bird** (a 40-triangle merged mesh with a flapping vertex-shader wing) for swarms of up to 120.

- Twelve species by palette: robin red, bluebird, goldfinch, cardinal, sparrow brown, dove white, starling black-green, jay blue-gray, wren tan, oriole orange, finch rose, owl gray.
- **Flock behavior:** simple boids (separation 1.0, alignment 0.6, cohesion 0.4) plus a target-seeking force toward the player or kite; cap neighbor checks with a spatial grid so 120 birds cost under 1 ms.

17. Audio Specification

17.1 Engine

- One `AudioContext`; nodes: `master` (gain) → `compressor` → destination. Buses: `music`, `sfx`, `amb`, `voice`, each a gain into `master`. Defaults: master 0.8, music 0.5, sfx 0.8, amb 0.6, voice 0.4.
- **Ducking:** when dialogue is active, lower `music` and `amb` to 60% over 0.2 s; restore afterwards.
- **Reverb:** one shared `ConvolverNode` with an impulse generated from decaying noise (length 1.8 s) and per-scene wet levels: dome 0.5, mansion halls 0.35, kitchen 0.1, lake 0.45, living room 0.12, outdoors 0.15.
- **Spatial:** `PannerNode` (HRTF, `inverse` distance, `refDistance` 1.5) for emitters; update the listener from the camera each frame.
- `Audio.unlock()` on the first click; all sounds are no-ops until unlocked. Everything is wrapped in try/catch so audio failure can never break the game.

17.2 Music sequencer

A tiny step sequencer (lookahead scheduling at 25 ms ticks, 100 ms ahead) that plays patterns of `[stepIndex, midiNote, durationSteps, velocity]` on named instruments. Instruments:

- **Music box:** sine + triangle one octave up, sharp attack, exponential decay 1.2 s, slight detune ± 8 cents.
- **Pump organ:** two detuned sawtooth oscillators through a lowpass (cutoff 900 Hz) with slow tremolo (5 Hz, depth 0.2).
- **Pad:** three detuned triangles with a slow filter sweep.
- **Bass:** a sine plus square an octave down, short envelope.
- **Flute:** a sine with breath noise (bandpassed white noise) and vibrato (5.5 Hz, depth 12 cents).

- **Percussion:** noise bursts for brush and snare, a low sine thump for kick, a high noise tick for hats.
- Add a global “wobble” (a slow pitch LFO ± 15 cents) to give a warped, memory-tape feel, scaled by the Flea Meter.

Chapter music table (key / tempo / instruments / character):

- **Title:** A minor, 72 BPM, music box + pad. Lullaby that never quite resolves.
- **Ch1:** TV jingle (invented, major, 108 BPM, plinky) that is cut dead on the TV hit; then silence and a ringing hum.
- **Ch2:** D minor, 110 BPM, organ + bass; frantic stabs while throwing, melting into a driving bass at the car.
- **Ch3:** E minor, 124 BPM, bass pulse + hats; the beat locks to engine RPM; a ghostly organ line enters when Elsa's voice surfaces.
- **Ch4:** C major to A minor, 60 BPM, pad + sparse music box; dawn birdsong over it.
- **Ch5:** no tempo: a held pad with slow harmonic drift, sine “moon hum” at 110 Hz with beating partials.
- **Ch6:** silence in the dark (breathing, house creaks); the lantern brings in a skittering music-box ostinato at 140 BPM; the chase adds percussion.
- **Ch7:** waltz in 3/4 at 84 BPM, music box + organ; the song mini-game plays its own melody; the transformation drops all but a single held note.
- **Ch8:** nearly silent: water lap, crickets, a faint pad; the fishing reel adds a slow heartbeat thump (60 BPM). No music for the first 60 s.
- **Ch9:** bright C major, 96 BPM, music box + organ; comic waltz during the Gronfiser verse.
- **Ch10:** a stately minor march at 88 BPM (organ + bass) that gains a nervous hat as the Flea Meter climbs.
- **Ch11:** a circus polka in D minor at 150 BPM that accelerates by +1 BPM per 4 seconds up to 172; flute joins during the dance; everything detunes as drunk rises.
- **Epilogue:** the title lullaby with one note missing.

17.3 SFX recipes (synthesis)

Keep each to a function `sfx.name(opts)` :

- **kick** : low-frequency thump (sine 140→40 Hz, 0.12 s) + cloth swish (noise burst, highpass 2 kHz).

- **crtPop** : a noise burst through a bandpass sweeping 6 kHz → 400 Hz plus a descending sine whine 12 kHz → 800 Hz over 0.6 s, then a faint 15.7 kHz fade.
- **glassShatter** : 8–12 short noise bursts with randomly high bandpass centers, staggered over 0.4 s.
- **splash** : lowpassed noise with a fast attack and a 0.5 s decay + 3 bubble blips (sine 300→900 Hz).
- **engine** : two sawtooth oscillators 55–140 Hz tracking RPM, lowpass 400 Hz + rumble noise; add a tire-hiss noise layer proportional to speed.
- **chirp** : a sine sweep 2.2→4.2 kHz in 60 ms with a quick amplitude envelope; variants by species (different base and rhythm).
- **wind** : bandpassed noise with slow sweep 200→900 Hz; gain follows the gust envelope.
- **vase** : a ceramic ring (two sines 1.8 kHz and 2.7 kHz, 0.3 s decay) + scatter ticks.
- **chop** : a wood “tok” (sine 180 Hz, 40 ms) + noise crack; **crack** for splits.
- **fire** : crackle (random short noise ticks, density by intensity) over a lowpassed roar.
- **flute(note)** : the flute instrument above with a breath attack.
- **bobber** : a soft plop (sine 520→220 Hz, 80 ms).
- **fishThrash** : filtered noise bursts with irregular timing.
- **grackle** : a rapid, creaky rising chirp with a ratchet LFO on a sawtooth, 0.5 s.
- **knife** : a short highpassed noise scrape.
- **gold** : a bright triple sine “ing” (1.2, 1.8, 2.4 kHz) used for pickups.
- **paper** : a filtered noise rustle.
- **thud** , **step** (alternating pitch, surface-dependent: wood, tile, dirt, carpet, grass), **door** , **lock** , **switch** .

17.4 Voice blips

Each character has a base frequency, waveform, blip length and pitch jitter; one blip per 2 characters of text (skip spaces), pitch nudged by punctuation:

- Christopher: square, 120 Hz base, 35 ms, ±15%.
- Elsa Ray: triangle, 230 Hz, 40 ms, ±10%.
- Mary Sue: sine, 380 Hz, 30 ms, ±12%.
- Frank: sine, 300 Hz, 45 ms, ±5% (steady, flat).
- The Sweaty Man: sawtooth lowpassed, 90 Hz, 50 ms, wobble.
- The Errante: sine + sub-octave, 70 Hz, 80 ms, heavy reverb.

- Clement: a single low sine, 60 Hz, 120 ms, only on “Hello, son.”
- The birds: chirps in a choral chord (three sine chirps at a major triad) timed to the syllables.

17.5 Ambience beds

Each scene starts a loop mix: Ch1 (fridge hum, clock tick, TV, wind); Ch3 (engine, wind rush, tire hiss, distant owl); Ch4 (crickets fading to birdsong, grass rustle); Ch5 (moon hum, tiny glass resonance); Ch6 (house creak, drip, heartbeat of the dark); Ch8 (water lap, crickets, frog, rare grackle); Ch9 (distant kitchen clatter, birds at windows); Ch11 (fire roar, night insects, owls, far-off wind).

18. UI Specification

18.1 Layers (z-order, bottom to top)

1. `canvas` (WebGL).
2. World-space DOM labels (price tags, nameplates), positioned each frame by projecting 3D points.
3. HUD (Flea jar, hearts, objective, notepad badge, FPS in debug).
4. Prompt and reticle.
5. Caption/dialogue box and choice buttons.
6. Full-screen overlays (the letter, the notepad, the chapter card, the flea crawl overlay, the fade).
7. Letterbox bars.
8. Menus (pause, settings, chapter select) and the loading card.

18.2 Visual style

- **Typography:** narration and captions in `Georgia, 'Palatino Linotype', serif` at `clamp(18px, 1.6vw, 26px)`; HUD in `ui-monospace, Menlo, Consolas, monospace` at 13–14 px; chapter titles in small caps with 0.12em letter spacing, 44–56 px, gold (`#d9b45a`) with a subtle text shadow.
- **Palette for UI:** parchment `#efe3c2`, ink `#2b1b12`, gold `#d9b45a`, ruby `#8a1d2b`, moon-silver `#cfd8ea`, dark panel `rgba(14,10,8,0.78)`.

- **Caption box:** bottom-center, width `min(900px, 92vw)`, padded 18–22 px, rounded 14 px, 1 px gold border, a slight paper-grain texture generated on canvas. Speaker tag above the box.
- **Choices:** vertical buttons, numbered, with a subtle gold underline on hover; the canon choice has no special mark (do not telegraph it).

18.3 HUD elements

- **Flea jar (top-left, 64×80 px):** a glass jar drawn with SVG or canvas; a fill level equal to `flea/100`; tiny dots (fleas) jiggle faster as it fills; the lid rattles above 80.
- **Household Hearts (top-right):** four small heart icons labeled by initial (E, M, F, C); each fills proportionally to its 0–10 value; a short pulse when it changes.
- **Objective (top-center):** one line, italic, fades to 40% opacity after 6 s unless a hint is active.
- **Notepad badge (bottom-right, visible from Chapter 10):** a small pad icon with a running total in gold digits.
- **Prompt:** centered below the reticle, e.g., `[E] Open the letter`.
- **Meter ticks:** small floating `+3 fleas` or `-1 ♥` that rise and fade near the meters.
- **Debug overlay (only with `debug=1`):** FPS, ms, draw calls (`renderer.info.render.calls`), triangles, scene name, chapter, flags, and buttons for “next chapter,” “add flea,” and “god mode.”

18.4 Screens

- **Title screen:** a slow flythrough of the dawn mansion on a rail with the moon (eye closed) hanging above; title in gold; buttons: Begin, Chapters, Settings, Credits. After 8 s idle, the moon’s eye opens and looks at the cursor. A line of rhyme below the title cycles every 6 s.
- **Chapter card:** black screen, the chapter number and title fade in (gold), then a single italic epigraph line (an original rhyme). Duration 3.5 s, skippable.
- **Pause menu:** Resume, Settings, Chapter Select, Controls, Quit to Title; background blurred with CSS `backdrop-filter` (falls back to a dark overlay).
- **Settings:** Quality (Auto/High/Low), Volume sliders (master, music, sfx), Mouse sensitivity, Invert Y, Reduced Motion, Subtitles (on by default; cannot be fully removed because dialogue is text), Text size (S/M/L), High-contrast HUD.
- **Letter UI (Chapter 1):** a large parchment panel with the letter text revealed line by line as the player scrolls or holds `Space`; the rhyme text is fully included.

- **Notepad UI (Q)**: a handwritten-style page (use a cursive-like system font stack such as 'Segoe Script', 'Bradley Hand', 'Comic Sans MS', cursive) listing items by room with gold values and a running total; new items scribble in with a stroke-reveal animation.

18.5 Accessibility and input

- **Reduced Motion**: disables camera shake, head bob, chromatic aberration, FOV breathing and the screen-edge flea crawl; replaces rapid flashes with fades.
- Never flash the screen faster than 3 times per second.
- Every mini-game has an **Assist** toggle in settings: longer timing windows ($\times 1.6$), slower speeds ($\times 0.75$), larger hit zones ($\times 1.3$), and optional auto-complete after 3 failures.
- All mouse-driven actions have keyboard alternatives; touch controls (if implemented) offer a virtual stick, a look-drag area and large action buttons.
- Subtitle text always has a dark panel for contrast. Text size and contrast follow the settings.